

# Laboratory 1

## K-Means Clustering

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# Overview of exercises

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- Laboratory 1: K-Means Clustering (1.0 Point)
- Laboratory 2: Grid Search for Linear Regression (1.0 Point)
- Laboratory 3: Gradient Descent for Linear Regression (1.5 Point)
- Laboratory 4: Logistic Regression (1.0 Point)
- Laboratory 5: Basics of Neural Network (1.0 Point)
- Laboratory 6: Image Classification with Neural Network (1.5 Points)
- Laboratory 7: Reinforcement learning (1.5 Point)
- Laboratory 8: Deep RL (1.5 Point)

# Dataset

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- A set of unlabeled training examples  $\{\mathbf{x}^{(i)}, i = 1, \dots, m\}$ ;  $\mathbf{x}^{(i)} \in \mathbb{R}^n$ ;

Number of data points  
(unlabeled data)  
(here is  $m = 4$ )

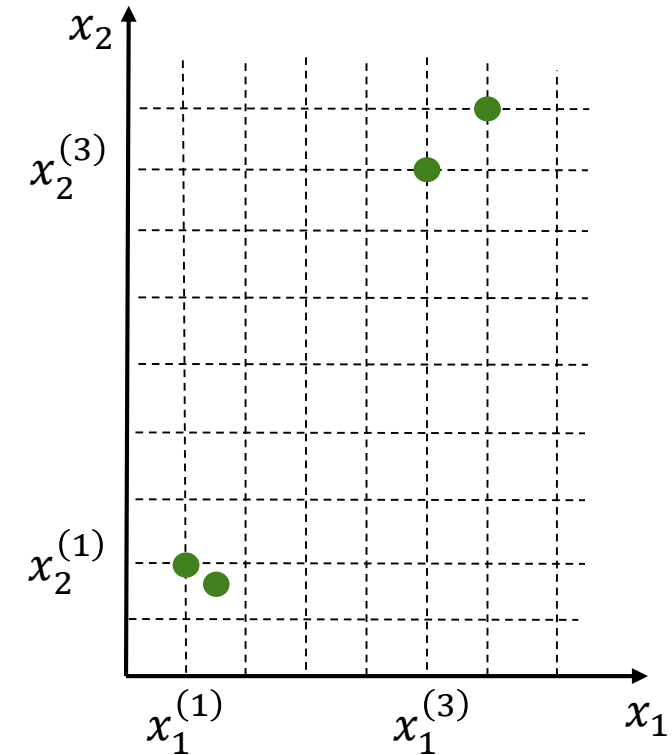
Dimension of points in each  
data point  
(here is  $n = 2$ )

# Dataset

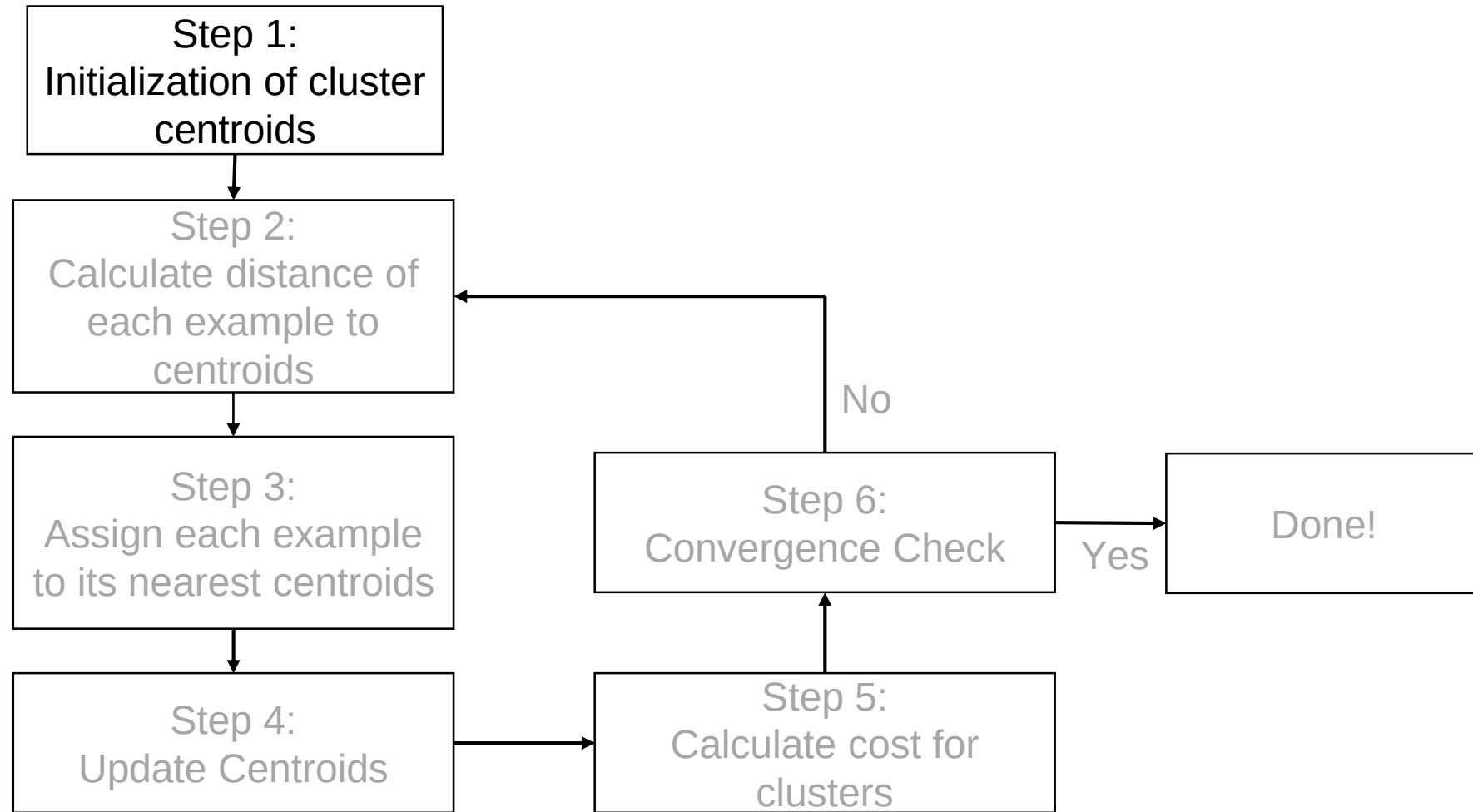
- Consider the following dataset in the table.

$$\mathbf{x}^{(i)} = [x_1^{(i)}, x_2^{(i)}], \quad i = 1, 2, 3, 4$$

| dataset            | $x_1^{(i)}$ | $x_2^{(i)}$ |
|--------------------|-------------|-------------|
| $\mathbf{x}^{(1)}$ | 1.0         | 2.0         |
| $\mathbf{x}^{(2)}$ | 1.5         | 1.8         |
| $\mathbf{x}^{(3)}$ | 5.0         | 8.0         |
| $\mathbf{x}^{(4)}$ | 6.0         | 9.0         |

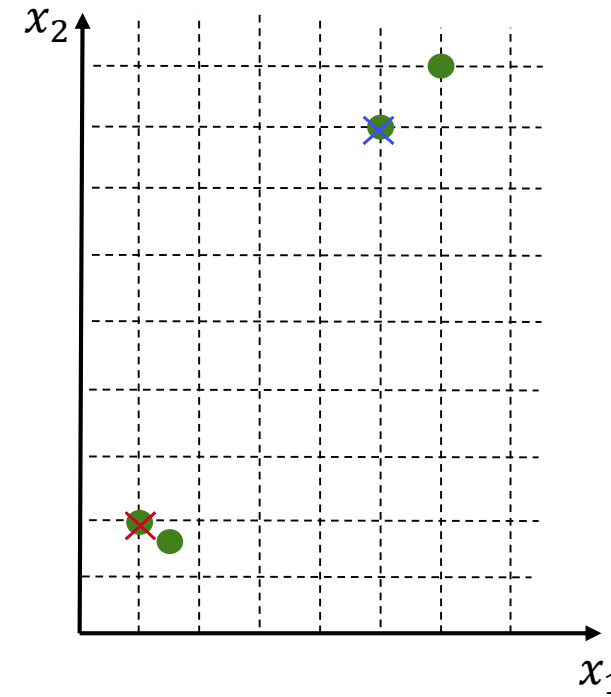


# K-means Clustering - Step 1

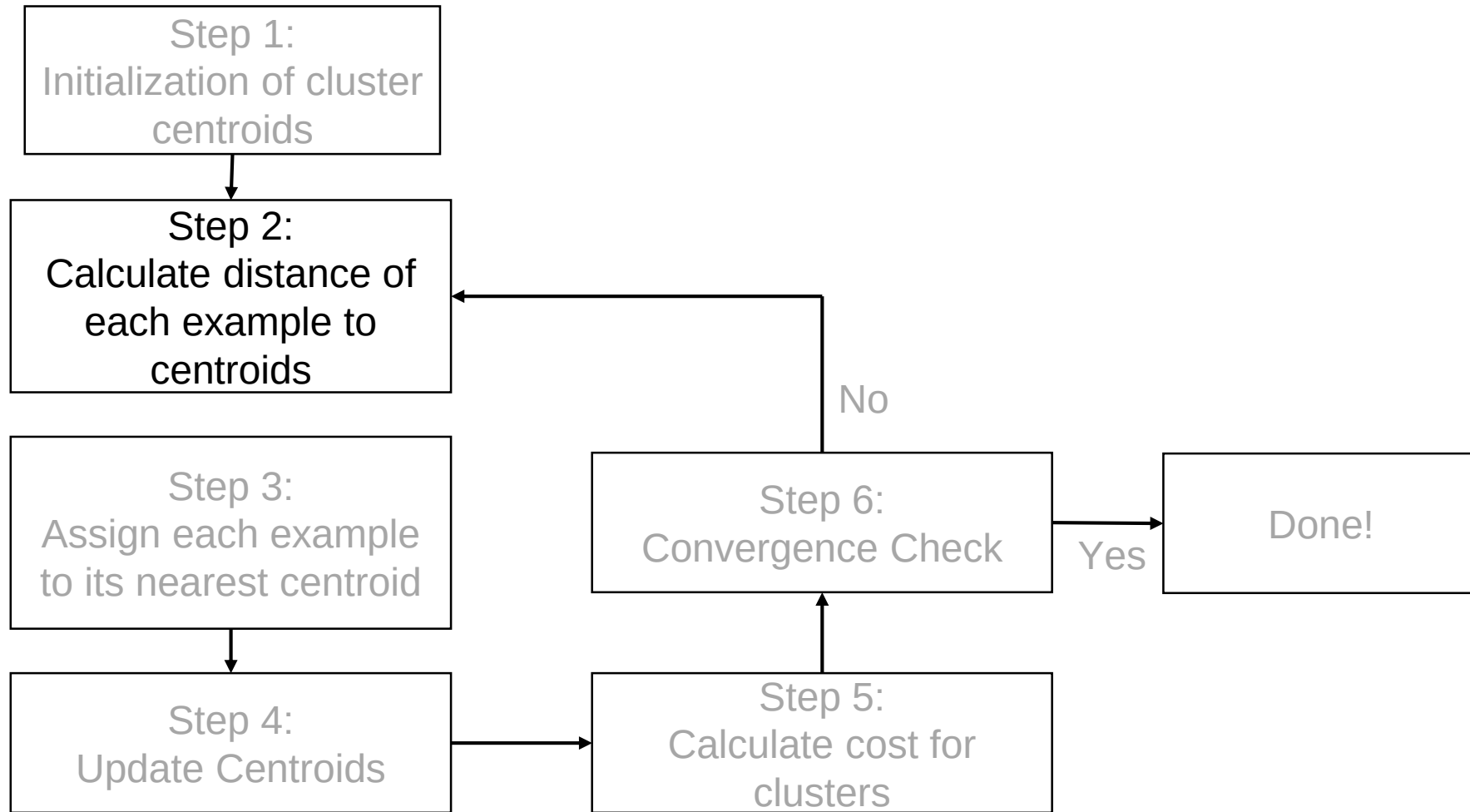


# Step 1: Random Initialization of Cluster Centroids

- Assume randomly initial centroids:
  - Centroid 1:  $\mu_1 = \mathbf{x}^{(1)} = (1.0, 2.0)$
  - Centroid 2:  $\mu_2 = \mathbf{x}^{(3)} = (5.0, 8.0)$



# K-means Clustering - Step 2





## Step 2: Calculating distance of each example to centroids

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- For every data example  $\mathbf{x}^{(i)}$ , squared Euclidean distance to each centroid is calculated.

$$d(\mathbf{x}^{(i)}, \boldsymbol{\mu}_k) = \|\mathbf{x}^{(i)} - \boldsymbol{\mu}_k\|^2$$

Where the number of data examples is denoted by  $i = 1, \dots, m$  and number of clusters is shown by  $k = 1, \dots, K$ .



## Step 2: Calculating distance of each example to centroids

- For the first training example, the distance from the first centroid is:

$$\begin{aligned}d(\mathbf{x}^{(1)}, \boldsymbol{\mu}_1) &= \|\mathbf{x}^{(1)} - \boldsymbol{\mu}_1\|^2 = (x_1^{(1)} - \mu_{1,1})^2 + (x_2^{(1)} - \mu_{1,2})^2 \\&= (1.0 - 1.0)^2 + (2.0 - 2.0)^2 = 0\end{aligned}$$

- The distance from the second centroid is:

$$\begin{aligned}d(\mathbf{x}^{(1)}, \boldsymbol{\mu}_2) &= \|\mathbf{x}^{(1)} - \boldsymbol{\mu}_2\|^2 \\&= (1.0 - 5.0)^2 + (2.0 - 8.0)^2 = 52\end{aligned}$$

| dataset            | $x_1^{(i)}$ | $x_2^{(i)}$ |
|--------------------|-------------|-------------|
| $\mathbf{x}^{(1)}$ | 1.0         | 2.0         |
| $\mathbf{x}^{(2)}$ | 1.5         | 1.8         |
| $\mathbf{x}^{(3)}$ | 5.0         | 8.0         |
| $\mathbf{x}^{(4)}$ | 6.0         | 9.0         |

## Step 2: Calculating distance of each example to centroids

- For the second example, the distance from the centroids are:

$$d(\mathbf{x}^{(2)}, \boldsymbol{\mu}_1) = (1.5 - 1.0)^2 + (1.8 - 2.0)^2 = 0.29$$

$$d(\mathbf{x}^{(2)}, \boldsymbol{\mu}_2) = (1.5 - 5.0)^2 + (1.8 - 8.0)^2 = 50.69$$

| dataset            | $x_1^{(i)}$ | $x_2^{(i)}$ |
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| $\mathbf{x}^{(1)}$ | 1.0         | 2.0         |
| $\mathbf{x}^{(2)}$ | 1.5         | 1.8         |
| $\mathbf{x}^{(3)}$ | 5.0         | 8.0         |
| $\mathbf{x}^{(4)}$ | 6.0         | 9.0         |

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$$d(\mathbf{x}^{(2)}, \boldsymbol{\mu}_2) = (1.5 - 5.0)^2 + (1.8 - 8.0)^2 = 50.69$$

- For the third example, the distance from the centroids are:

$$d(\mathbf{x}^{(3)}, \boldsymbol{\mu}_1) = (5.0 - 1.0)^2 + (8.0 - 2.0)^2 = 52$$

$$d(\mathbf{x}^{(3)}, \boldsymbol{\mu}_2) = (5.0 - 5.0)^2 + (8.0 - 8.0)^2 = 0$$

| dataset            | $x_1^{(i)}$ | $x_2^{(i)}$ |
|--------------------|-------------|-------------|
| $\mathbf{x}^{(1)}$ | 1.0         | 2.0         |
| $\mathbf{x}^{(2)}$ | 1.5         | 1.8         |
| $\mathbf{x}^{(3)}$ | 5.0         | 8.0         |
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$$d(\mathbf{x}^{(3)}, \boldsymbol{\mu}_1) = (5.0 - 1.0)^2 + (8.0 - 2.0)^2 = 52$$

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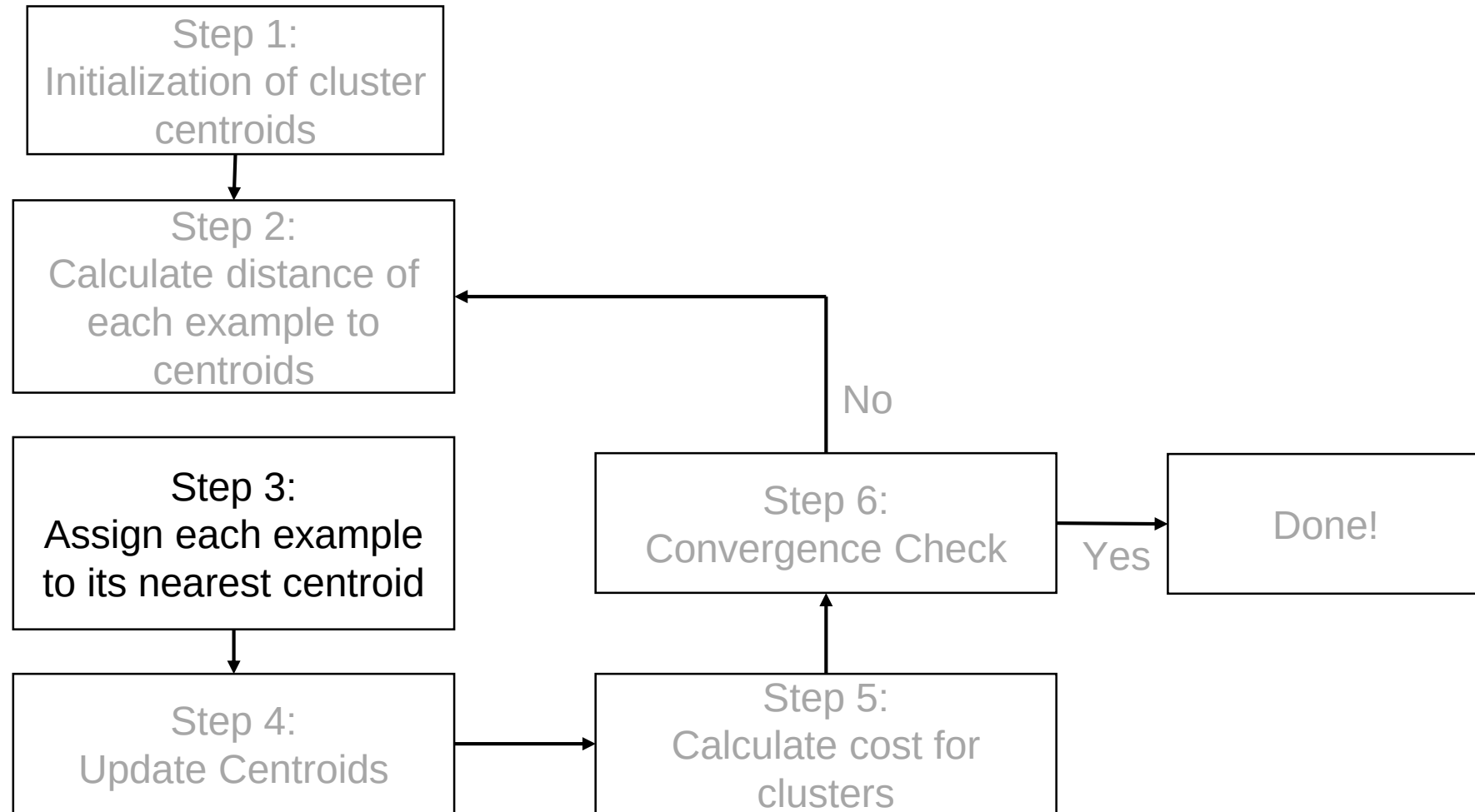
- For the fourth example, the distance from the centroids are:

$$d(\mathbf{x}^{(4)}, \boldsymbol{\mu}_1) = (6.0 - 1.0)^2 + (9.0 - 2.0)^2 = 74$$

$$d(\mathbf{x}^{(4)}, \boldsymbol{\mu}_2) = (6.0 - 5.0)^2 + (9.0 - 8.0)^2 = 2$$

| dataset            | $x_1^{(i)}$ | $x_2^{(i)}$ |
|--------------------|-------------|-------------|
| $\mathbf{x}^{(1)}$ | 1.0         | 2.0         |
| $\mathbf{x}^{(2)}$ | 1.5         | 1.8         |
| $\mathbf{x}^{(3)}$ | 5.0         | 8.0         |
| $\mathbf{x}^{(4)}$ | 6.0         | 9.0         |

# K-means Clustering - Step 3



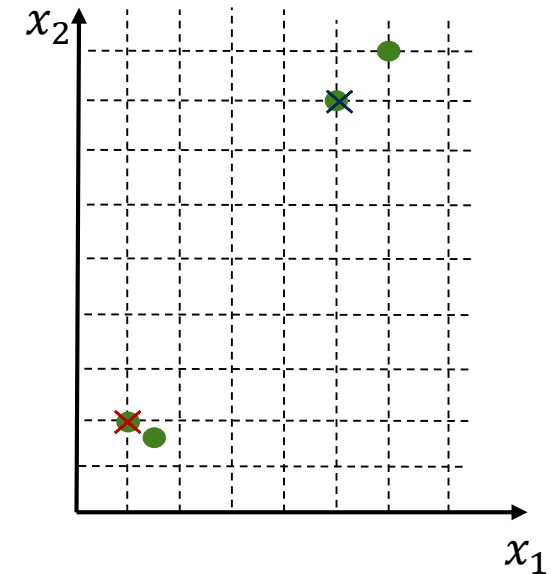
## Step 3: Assign Every Data example to its Nearest Cluster

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- Then every datapoint  $\mathbf{x}^{(i)}$  is assigned to the nearest cluster:

$$y^{(i)} = \arg \min_k d(\mathbf{x}^{(i)}, \boldsymbol{\mu}_k)$$

Here  $i = 1, 2, 3, 4$  and  $k = 1, 2$

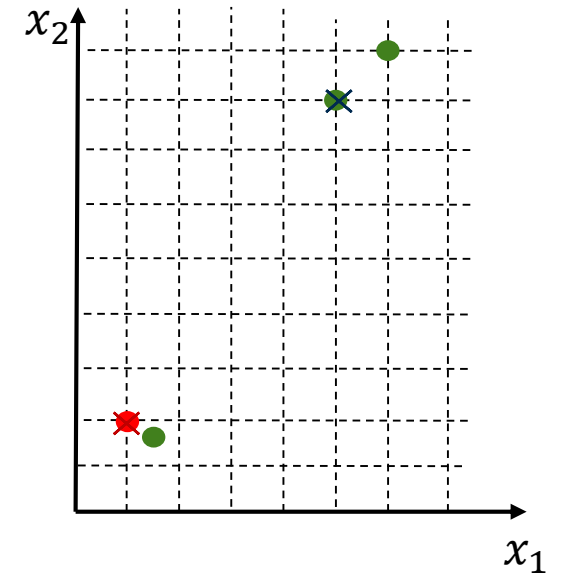


## Step 3: Cluster assignment

- Assign first data example to the nearest centroid's cluster.

$$y^{(1)} = \underset{k}{\operatorname{argmin}} [d(\mathbf{x}^{(1)}, \boldsymbol{\mu}_k)] = 1$$

Therefore,  $\mathbf{x}^{(1)}$  will be assigned to  $\boldsymbol{\mu}_1$ .



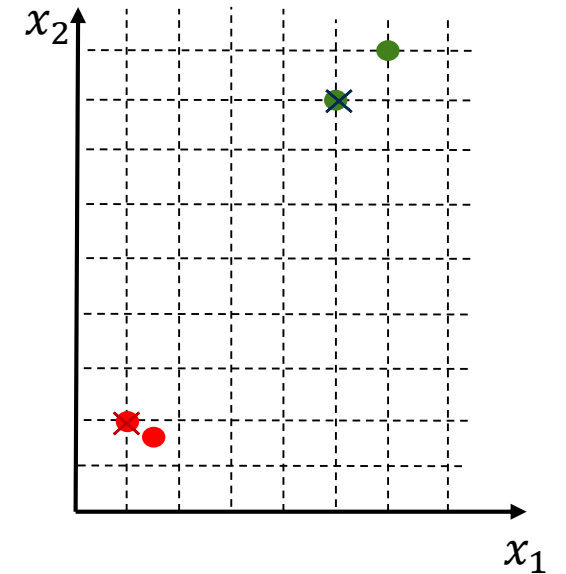


## Step 3: Cluster assignment

- Assign second data example to the nearest centroid's cluster.

$$y^{(2)} = \underset{k}{\operatorname{argmin}} [d(\mathbf{x}^{(2)}, \boldsymbol{\mu}_k)] = 1$$

Therefore,  $\mathbf{x}^{(2)}$  will be assigned to  $\boldsymbol{\mu}_1$ .

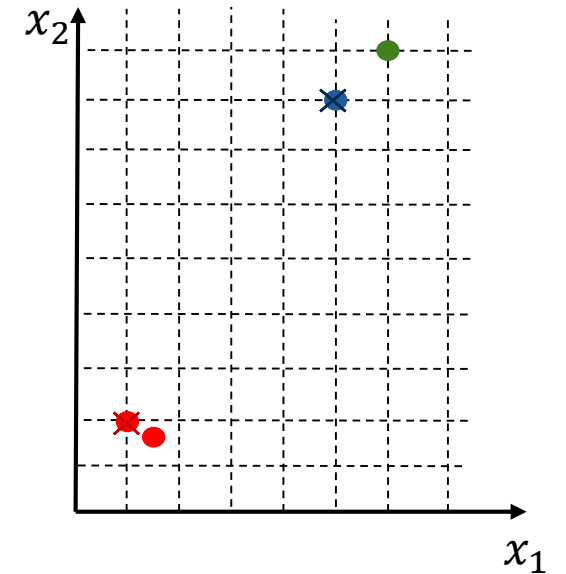


## Step 3: Cluster assignment

- Assign third data example to the nearest centroid's cluster.

$$y^{(3)} = \underset{k}{\operatorname{argmin}} [d(\mathbf{x}^{(3)}, \boldsymbol{\mu}_k)] = 2$$

Therefore,  $\mathbf{x}^{(3)}$  will be assigned to  $\boldsymbol{\mu}_2$ .

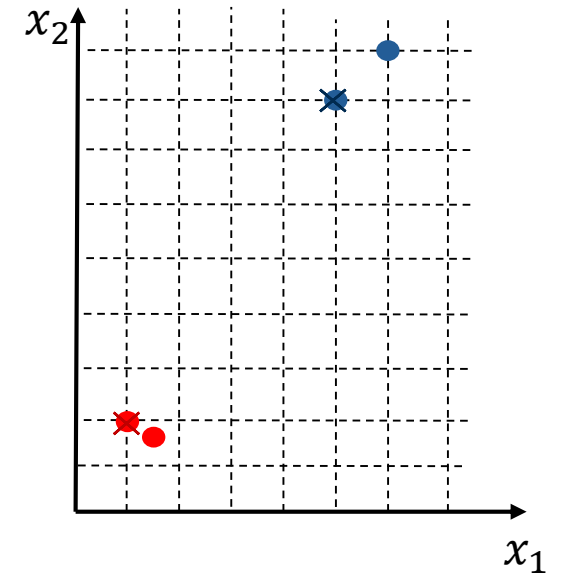


## Step 3: Cluster assignment

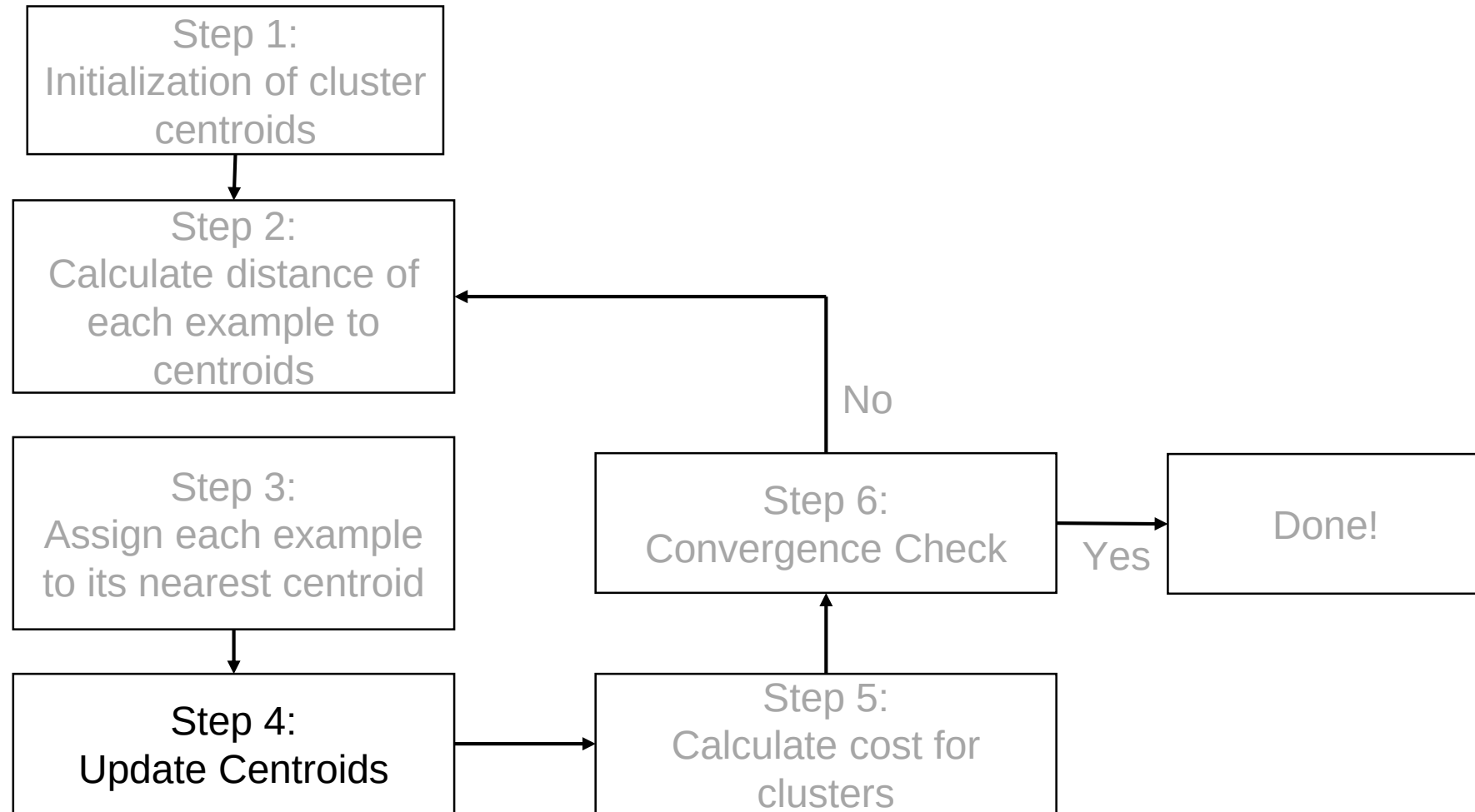
- Assign fourth data example to the nearest centroid's cluster.

$$y^{(4)} = \underset{k}{\operatorname{argmin}} [d(\mathbf{x}^{(4)}, \boldsymbol{\mu}_k)] = 2$$

Therefore,  $\mathbf{x}^{(4)}$  will be assigned to  $\boldsymbol{\mu}_2$ .



# K-means Clustering - Step 4

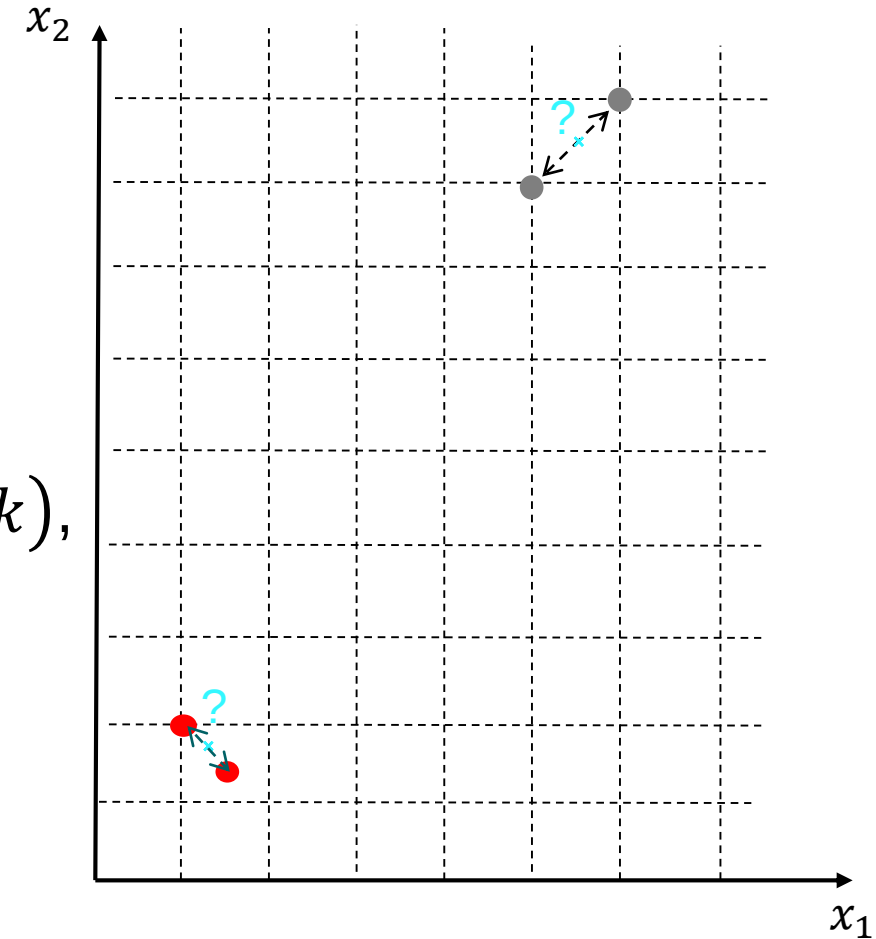


## Step 4: Update Centroids

- Update centroids:

$$\boldsymbol{\mu}_k = \frac{\sum_{i=1}^m 1(y^{(i)} = k) \mathbf{x}^{(i)}}{\sum_{i=1}^m 1(y^{(i)} = k)} \quad k = 1, \dots, K$$

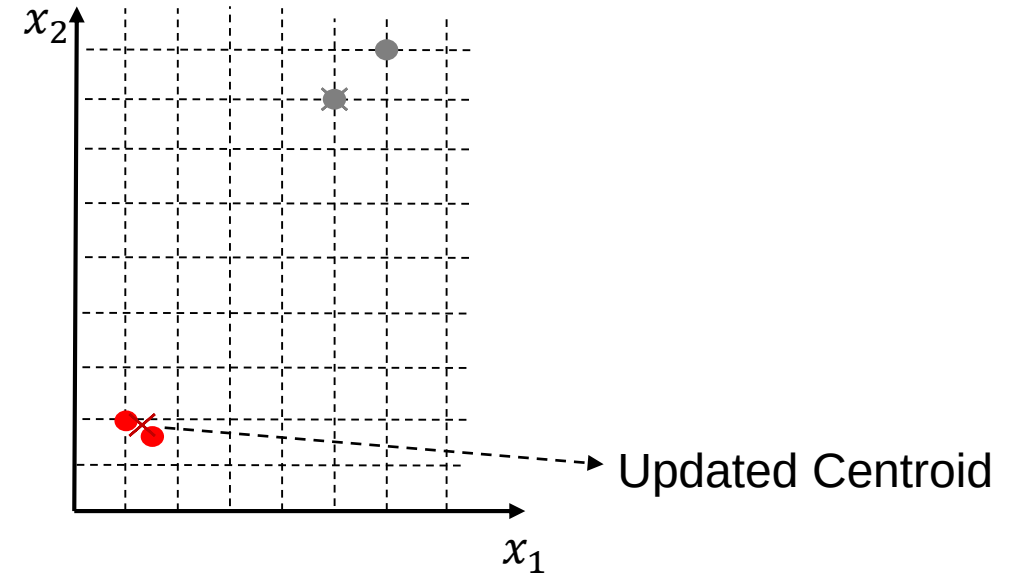
- Here,  $1(y^{(i)} = k)$  is the indicator function. It returns one when its argument is true ( $y^{(i)} = k$ ), otherwise (if  $y^{(i)} \neq k$ ), it returns zero.



## Step 4: Update Centroids

- Update  $\mu_1$  as a mean of  $\mathbf{x}^{(1)}$  and  $\mathbf{x}^{(2)}$

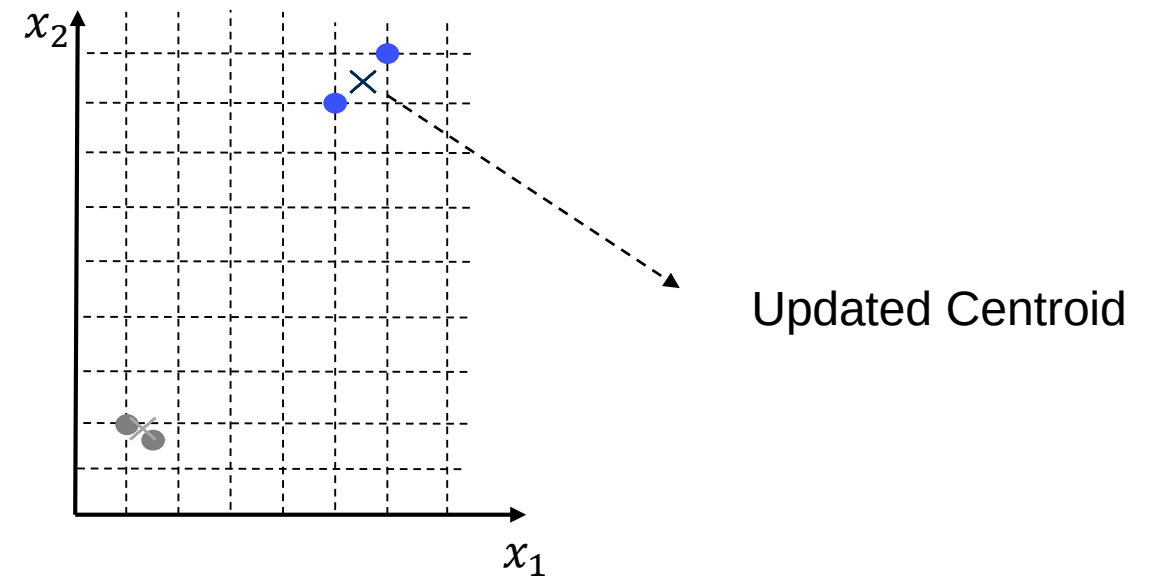
$$\mu_1 = \left( \frac{1.0 + 1.5}{2}, \frac{2.0 + 1.8}{2} \right) = (1.25, 1.9),$$



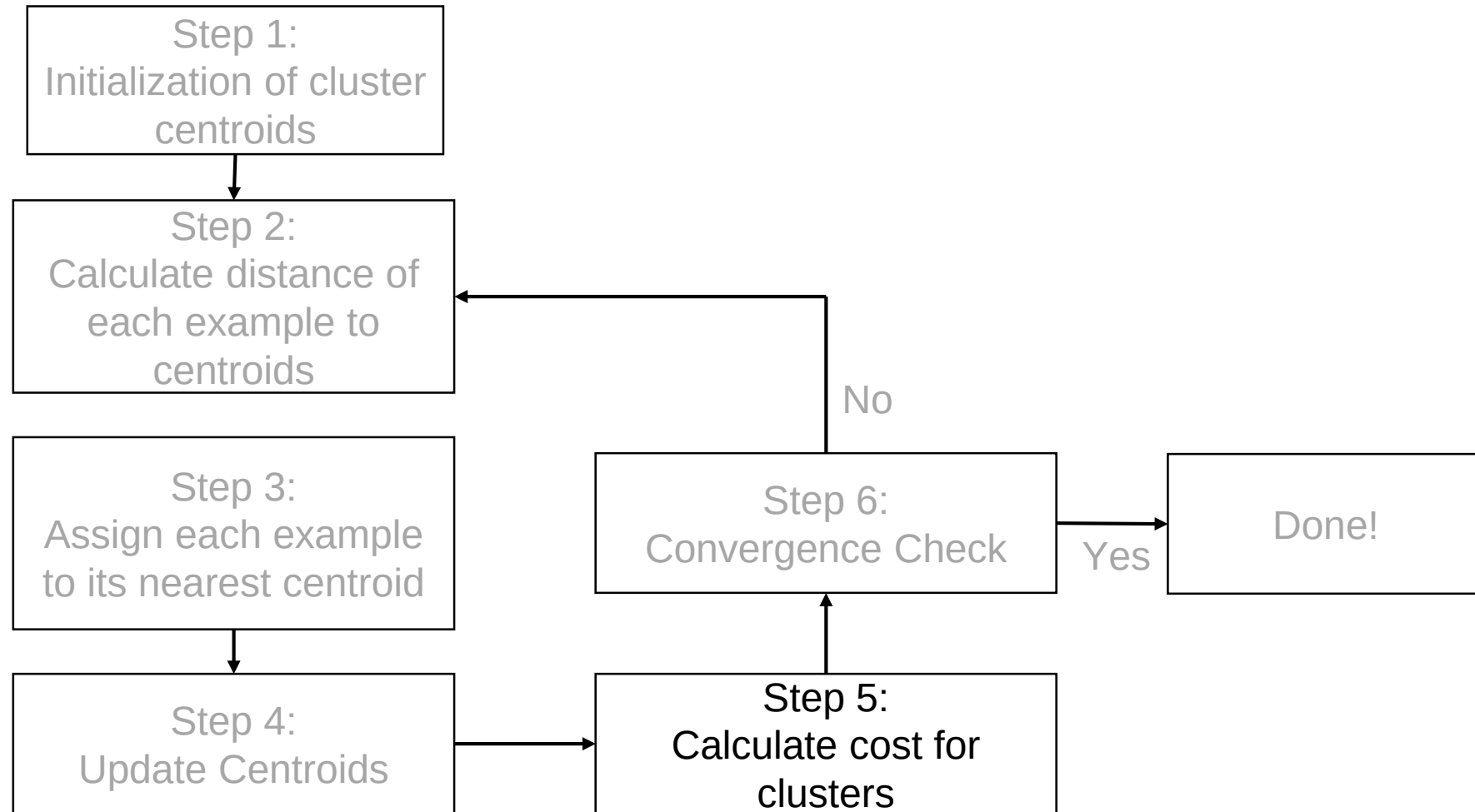
## Step 4: Update Centroids

- Update  $\mu_2$  as a mean of  $\mathbf{x}^{(3)}$  and  $\mathbf{x}^{(4)}$

$$\mu_2 = \left( \frac{5.0 + 6.0}{2}, \frac{8.0 + 9.0}{2} \right) = (5.5, 8.5).$$



# K-means Clustering - Step 5





## Step 5: Calculate cost for the cluster assignment

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$$J = \frac{1}{2m} \sum_{i=1}^m \left\| \mathbf{x}^{(i)} - \boldsymbol{\mu}_{y^{(i)}} \right\|^2 ,$$

- $y^{(i)}$  is the assigned cluster for  $\mathbf{x}^{(i)}$ , and  $\boldsymbol{\mu}_{y^{(i)}}$  is its centroid.

## Step 5: Calculate the Cost

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- Calculating cost value for the cluster assignment:

$$\begin{aligned} J(\boldsymbol{\mu}_1, \mathbf{x}^{(i)}) &= \frac{1}{2m} \sum_{i=1}^2 \left\| \mathbf{x}^{(i)} - \boldsymbol{\mu}_{y^{(i)}} \right\|^2 \\ &= \frac{1}{8} \left( (1.0 - 1.25)^2 + (2.0 - 1.9)^2 + (1.5 - 1.25)^2 \right. \\ &\quad \left. + (1.8 - 1.9)^2 \right) \\ &= 0.018 \end{aligned}$$

## Step 5: Calculate the Cost

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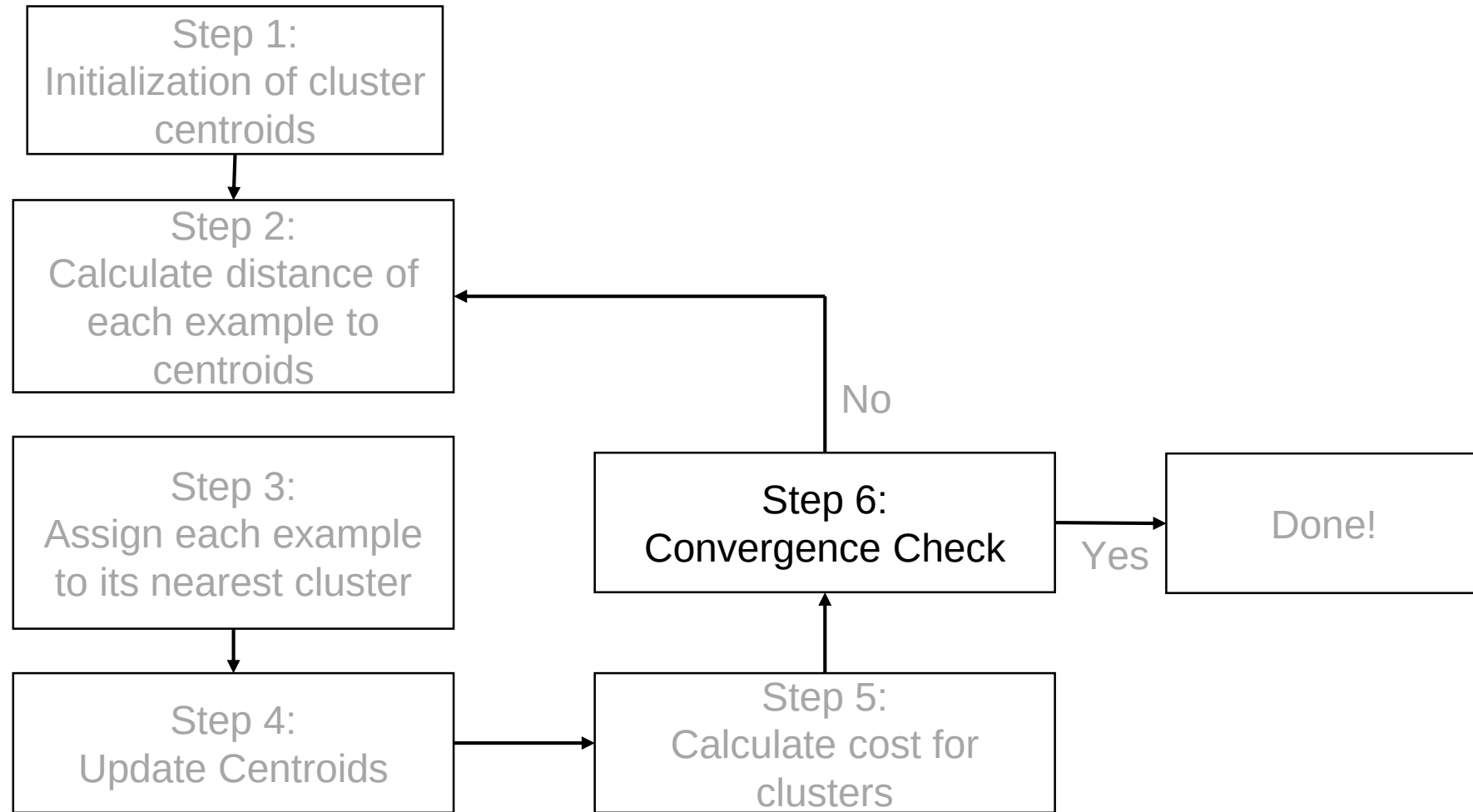
- Calculating cost value for the cluster assignment of points to  $\mu_2$ :

$$\begin{aligned} J(\mu_2, \mathbf{x}^{(i)}) &= \frac{1}{2m} \sum_{i=3}^4 \left\| \mathbf{x}^{(i)} - \mu_{y(i)} \right\|^2 \\ &= \frac{1}{8} \left( (5.0 - 5.5)^2 + (8.0 - 8.5)^2 + (6.0 - 5.5)^2 + (9.0 - 8.5)^2 \right) \\ &= 0.125 \end{aligned}$$

- Total cost:

$$J = J(\mu_1, \mathbf{x}^{(i)}) + J(\mu_2, \mathbf{x}^{(i)}) = (0.018 + 0.125) = 0.143$$

# K-means Clustering - Step 6



## Step 6: Convergence Check

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- Repeat Steps 2. to 5. until the convergence criteria is met.
- The convergence criteria is when the centroids are not updated anymore (or move slightly) or when the cost value does not change, or the maximum number of iterations is met.



# Assignment

# K-Means clustering for Iris dataset

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- **Objective:** Use the Iris dataset and cluster the data into  $K$  clusters. For this assignment follow the steps mentioned earlier, choose cluster centroids, calculate the cost for the clusters and update the cluster centroids to reach the convergence criteria.
- Visualize the error convergence (cost) over iterations.

# Iris dataset

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- **Dataset:** The Iris dataset is readily available in the *sklearn.datasets* module. This dataset consists of 150 samples of iris flowers, each described by four features: sepal length, sepal width, petal length, and petal width.
- There are three different classes ( $K = 3$ ) in the dataset, representing three different species of iris flowers.
- Loading the dataset: Importing the Iris dataset from the *sklearn.datasets* module. (No need for downloading or additional processing.)
- ```
from sklearn.datasets import load_iris  
# Load dataset  
data = load_iris()  
X = data.data
```



# Tasks

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**Task1.** Implement the  $K$ -means algorithm, as follows:

**Step 1.** Randomly select three data points from the dataset as initial cluster centroids.

**Step 2.** For each data point, calculate the squared Euclidean distance to each centroid.

**Step 3.** Assign each data point to the nearest centroid's cluster based on the minimum calculated distance.

**Step 4.** Recalculate the centroids for each cluster by taking the mean of all data points assigned to that cluster.

**Step 5.** Calculate the cost value

## Tasks, contd.

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**Step6.** Convergence Check: Repeating Steps 2. to 5. until the change in centroids is smaller than  $\epsilon = 1 \times 10^{-6}$ .

**Task2.** Multiple Runs: Re-run the *K*-means algorithm 10 times with different initial centroids to find the clustering result with the minimum cost value.

**Task3.** Plot the cost over iterations for the best run to visualize convergence.

# Submission

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- Functions you need to implement (Rest of the code is given):
  - *initilize\_centroids(x, K, seed=None)*
  - *calculate\_distances(a, b)*
  - *assign(x, c)*
  - *move\_centroids(x, y, k, seed=None)*
  - *cost(x, c, y)*
- Submit your Python code as .py file.
  - If you experience any issues uploading the .py files directly, compress it into a .zip or .tar file and upload the compressed file instead
- If the code fails to run, it will receive no points.





# End of Exercise 1